Supplement V4.1

Original-world grouped statistical and artifact audit

This supplement accompanies **“Organizational-memory claims under original-world grouped validation: a V4.1 reconciliation.”** It reports methods, sample sizes, fold definitions, audit findings, and numerical dispositions in greater detail. No new simulation was run.

S1. Canonical-source chain

The exact V4.0 source is Git commit `23b53aee4169deeda30aad2a9dba8587024f8d3d`, subject `CANONICAL-REANALYSIS-V4: adopt committed reproducible headline numbers`. The release manifest is `v4.0/canonical/release/RELEASE_MANIFEST_V4.json`.

The canonical primary objects are:

Object	Canonical path
Manuscript source	`docs/paper/full/ORGANIZATIONAL_MEMORY_FULL_MANUSCRIPT_V4.tex`
Manuscript PDF	`docs/paper/full/ORGANIZATIONAL_MEMORY_FULL_MANUSCRIPT_V4.pdf`
Supplement source	`docs/paper/full/SUPPLEMENT_V4.tex`
Supplement PDF	`docs/paper/full/SUPPLEMENT_V4.pdf`
Claim ledger	`docs/paper/full/CLAIM_LEDGER_V4.md`
Statistical re-audit	`docs/paper/full/STATISTICAL_REAUDIT_V4.md`
Reproduction package	`reproduction/`
Longitudinal raw data	`results/observer/tca_holdout_raw.pkl`

All 37 release-manifest files verify byte-for-byte against their declared SHA-256 hashes. V4.0 is not edited by V4.1.

S2. Analysis units and sample sizes

S2.1 Primary longitudinal artifact

- rows: n=36 at each checkpoint;
- original worlds: W=3, seeds 38502–38504;
- imposed histories: H=12;
- rows per world: 12;
- checkpoints: initial, moderate, deep-1, and deep;
- outer group: original world / simulation seed;
- response coordinates: cumulative drive h1 and order contrast h2;
- primary component rule: longitudinal tracker;
- sensitivity component rule: largest component at each frame.

S2.2 Common-recipient transplant artifact

- rows: n=36;

- donor worlds: W=3, seeds 35001–35003;
- histories: H=12;
- recipient bodies: one fixed standardized recipient;
- response vector: the committed five-dimensional response;
- outer group: donor world.

The analysis therefore addresses donor-world generalization for one recipient, not recipient-world generalization.

S2.3 In-place response artifact

- rows: n=24;
- original worlds: W=2;
- histories: H=12;
- status: descriptive only because W=2.

S2.4 Historical body-baseline artifact

- rows: n=48;
- original worlds: W=4, seeds 38501–38504;
- histories: H=12;
- component rule: largest-component reselection;
- compared feature sets: component-memory features and size-plus-mass.

S3. V4.0 bootstrap audit

The V4.0 implementation draws H=12 history identifiers with replacement from the same 12 histories. It copies the associated rows and assigns each draw a fresh group identifier before leave-one-group-out fitting.

If a history is selected twice, its copied rows are exact duplicates, including the same original-world observations. The fresh identifiers make the copies appear independent to the cross-validation splitter. Consequently, one copy can be in a training fold while another exact copy is in the corresponding test fold.

The probability of no duplicate in one replicate is:

`P(all unique) = 12! / 12^12 = 0.0000537232`.

The probability of at least one duplicate is:

`P(duplicate) = 0.9999462768`.

Audit of the fixed V4.0 draws gives:

Unique histories in replicate	Number of replicates
4	3
5	43
6	291
7	854
8	1,081
9	593
10	122
11	13
12	0

Thus 3,000/3,000 realized replicates contained duplicates and 0/3,000 were leakage-free under the intended original-observation interpretation.

S4. Frozen V4.1 prediction algorithm

For response y, feature matrix X, and original-world vector g:

1. enumerate each unique world w; 2. define test rows as g=w and training rows as g≠w; 3. identify nonconstant columns using training rows only; 4. estimate feature mean and standard deviation using training rows only; 5. standardize training and test rows with those training estimates; 6. fit ridge regression with λ=1 and an intercept; 7. predict all rows in world w once; 8. concatenate all held-out predictions in original row order.

No original world appears in both training and test within an outer fold. Preprocessing is nested inside the fold. No hyperparameter is selected after outcomes are inspected.

The primary pooled score is:

`R² = 1 - sum_i (y_i - yhat_i)^2 / sum_i (y_i - mean(y))^2`.

For uncertainty display, each fold score uses a normalized held-out loss with a common outcome scale, so the mean fold score equals the pooled score.

S5. Crossed world-and-history exclusion

The primary estimator tests generalization to a new world while retaining other worlds' observations of the same imposed history in training. A stricter sensitivity removes both:

- every row from the held-out world; and
- every training row with the test row's history identifier.

This produces one prediction per test row from other worlds and other histories. It is a sensitivity analysis because its training set differs by test history.

Checkpoint	h1 primary	h1 crossed	h2 primary	h2 crossed
Initial	0.7382	0.6021	0.7631	0.6618
Moderate	0.9020	0.8842	-0.5102	-0.9873
Deep-1	0.8432	0.8041	-0.1315	-0.4677
Deep	0.6947	0.6420	-1.1183	-1.4230

S6. Original-world fold scores and uncertainty

S6.1 Cumulative-drive coordinate h1

Checkpoint	World 38502	World 38503	World 38504	Mean / pooled
Initial	0.7735	0.5479	0.8933	0.7382
Moderate	0.9006	0.8557	0.9498	0.9020
Deep-1	0.8845	0.7079	0.9373	0.8432
Deep	0.7454	0.4141	0.9246	0.6947

S6.2 Order-contrast coordinate h2

Checkpoint	World 38502	World 38503	World 38504	Mean / pooled
Initial	0.7488	0.6936	0.8470	0.7631
Moderate	-0.0162	-1.2059	-0.3085	-0.5102

Checkpoint	World 38502	World 38503	World 38504	Mean / pooled
Deep-1	0.2850	-0.9119	0.2324	-0.1315
Deep	-0.0982	-3.2362	-0.0206	-1.1183

S6.3 Descriptive intervals

Quantity	World-fold t interval	Fixed-fold block percentiles
h1 initial	[0.3026, 1.1738]	[0.5968, 0.8673]
h1 moderate	[0.7851, 1.0189]	[0.8654, 0.9391]
h1 deep-1	[0.5448, 1.1416]	[0.7462, 0.9258]
h1 deep	[0.0513, 1.3381]	[0.4859, 0.8858]
h2 initial	[0.5701, 0.9561]	[0.7056, 0.8257]
h2 moderate	[-2.0503, 1.0299]	[-1.0115, -0.0796]
h2 deep-1	[-1.8117, 1.5487]	[-0.6640, 0.2736]
h2 deep	[-5.6757, 3.4390]	[-2.5563, -0.0374]

With W=3, neither interval family is a reliable nominal 95% confidence interval. The fold values are the primary uncertainty report.

S7. Tracker and event-evidence audit

The longitudinal raw summary directly contains:

Count or quantity	Reproduced value
total records	36
recorded surviving	36
recorded lost	0
recorded switches	0
mean deep M	0.190203
deep M≤0.25	34/36

The deep h1 tracker comparison is:

World	Longitudinal	Largest each frame	Difference
38502	0.7454	0.7928	-0.0474
38503	0.4141	0.3722	0.0419
38504	0.9246	0.8470	0.0777
Mean	0.6947	0.6706	0.0241

The paired descriptive t interval for the mean difference is [-0.1360, 0.1841].

The persisted artifact does **not** contain:

- component masks by frame;
- candidate association edges;
- individual geometry/size gate terms;
- alternative compatible matches;
- split or merge adjudication records;

- a dedicated fusion flag.

Accordingly, the recorded survival and switch fields are reproducible, but fusion-free continuity is not independently reconstructable.

S8. Secondary results

S8.1 Common-recipient transplant

Response	Grouped R^2	World folds	Crossed result
full h1	0.6468	0.6568, 0.6972, 0.5863	0.4807
inert h1	0.0000	0, 0, 0	-0.1901
erased h1	0.0000	0, 0, 0	-0.1901
full h2	0.3537	0.4008, 0.4318, 0.2284	0.1002

The grouped constant-control score is zero because the held-out predictions equal the no-information mean under the primary fold normalization. The crossed row-wise sensitivity retains the historical negative value because its training means differ by excluded history. Neither representation contains positive information.

S8.2 In-place response

Response	Grouped R^2	Worlds	Disposition
h1 difference	0.7039	2	descriptive only
h2 difference	0.1939	2	descriptive only

S8.3 Component-memory versus body baseline

Checkpoint	Memory features	Size plus mass	Paired difference
Initial	0.8291	-0.1078	0.9369 [0.4994, 1.3744]
Deep	0.7692	0.4538	0.3154 [-0.2394, 0.8701]

The intervals are descriptive t intervals over W=4 original worlds. The deep comparison is not separated.

S9. Global information versus local storage

The primary labels are functions of a drive history applied to the complete world. Every large component in one world is therefore exposed to the same label. A successful component-feature decoder can establish access to global history-related information. It cannot, without additional frozen comparisons, show that the information is uniquely or specifically stored inside the tracked component.

The required but absent comparison would hold folds, preprocessing, labels, and model fixed while varying access to:

- tracked-component features;
- nearby-component features;
- target-masked environmental features;
- whole-world aggregate features.

No such local/neighbour/environment/world access matrix is committed in V4.0. The local-storage claim is withdrawn rather than inferred from tracker agreement.

S10. Figure-caption and sample-size audit

Figure	Artifact family	n	W	H	Caption disposition
Figure 1	longitudinal	36 per checkpoint	3	12	corrected; intervals labelled descriptive
Figure 2	tracker sensitivity	36	3	12	corrected; event-evidence limitation stated
Figure 3 transplant	common recipient	36	3 donor	12	corrected; one recipient stated
Figure 3 baseline	historical H2-CERT	48	4	12	corrected; different family and tracker stated

No figure caption treats rows or histories as independent worlds. No caption uses the withdrawn V4.0 bootstrap interval.

S11. Claim dispositions

ID	Short claim	V4.1 status
C01	Initial h1 global informational content	SUPPORTED
C02	Positive deep h1 point	SUPPORTED
C03	Deep h1 lower bound clears 0.50	NOT ESTABLISHED
C04	Initial h2 information	SUPPORTED
C05	Deep h2 information persists	NOT ESTABLISHED
C06	Tracker points are descriptively comparable	OBSERVED
C07	36/36 surviving and zero switches are recorded	OBSERVED
C08	Fusion-free continuity independently verified	WITHDRAWN
C09	Common-recipient response carries h1 information	SUPPORTED
C10	In-place response inferentially established	NOT ESTABLISHED
C11	Deep memory features outperform size-plus-mass	NOT ESTABLISHED
C12	Tracked component is the local storage site	WITHDRAWN

The authoritative detailed wording is in `CLAIM_LEDGER_V4_1.md` and `claim_ledger_v4_1.json`.

S12. Reproduction boundary

The V4.1 analysis reads only four committed pickle artifacts. It does not instantiate the simulation engine, execute a seed, replay a trajectory, or reconstruct a frame. The exact commands are listed in `REPRODUCIBILITY.md`.

The unsealed turnover and active-reconstruction programme is outside this paper. No result from that programme appears in the manuscript abstract, results, or conclusion.